FED4FED: Statistical Fairness Evaluation Report

Privacy-Preserving Verification for Federated Learning Models

FAIRNESS VERDICT: UNFAIR (Statistical Disparity Detected)

Under the Conservative Joint Policy (Disparity detected if either Alg 1 or Alg 2 rejects H0), the global federated model exhibits statistically significant performance disparities across clients at alpha = 0.05. Alg 1 p-value: 6.0337e-34, Alg 2 p-value: 7.9731e-31.

1. Experiment & Architecture Configuration

Experiment ID:	mnist_unfair_20261020_222651	**Date / Time:**	2026-10-20 22:32:27
Dataset:	MNIST	**Scenario:**	Unfair
Clients (K):	15	**Seed:**	42
FL Rounds:	10	**Final Val Accuracy:**	0.8479
Partition Method:	clustered	**Bootstrap Replications (B):**	350

2. Statistical Fairness Hypothesis Tests

Test Algorithm	Statistic	df	p-value	Significance (α)	Decision
Algorithm 1 (Graybill-Deal / T1)	194.9003	14	6.0337e-34	0.050	**Reject H0 (Disparity)**
Algorithm 2 (Welch-Type F)	13.8161	(14, 1136.6)	7.9731e-31	0.050	**Reject H0 (Disparity)**

3. Algorithm 3: Global Performance Confidence Interval

Status: UNAVAILABLE (Condition Not Met)
Decision Rule Rationale: Algorithm 2 (Welch-Type F Test) rejected the null hypothesis of equal performance (p = 7.9731e-31 < alpha = 0.05). Because performance disparity is statistically significant across clients, the common mean assumption mu_1 = ... = mu_K does not hold. A pooled global confidence interval cannot be constructed under Algorithm 3 decision rules.

4. Sample Decentralized Client Performance Statistics

Client ID	Test Samples (n_i)	Observed Metric	Bootstrap Mean	Variance (S_i^2)	Std Error (SE)
Client 00	666	0.8138	0.8153	0.000213	0.0146
Client 01	667	0.8876	0.8876	0.000134	0.0116
Client 02	668	0.7874	0.7890	0.000277	0.0167

Client 03	666	0.8964	0.8956	0.000126	0.0112
Client 04	667	0.7541	0.7539	0.000290	0.0170
Client 05	667	0.8726	0.8735	0.000188	0.0137
Client 06	666	0.8694	0.8686	0.000142	0.0119
Client 07	666	0.8679	0.8684	0.000181	0.0135

Scientific Methodology: Fed4Fed privacy-preserving statistical homogeneity testing using non-parametric bootstrapping, Graybill-Deal common mean estimation, and Welch-type heteroscedastic F-testing. Generated by FED4FED Research Suite v1.0.0.